Probability and Distributions Transcript

Hi everyone, today we are going to learn about probability distributions, but before we get into that, let me walk you through what we have understood till now. We went through random variables, which basically is a concept of attaching a numerical value to any random event or trial, right. So let me give you a walk around of what exactly are random variables, we have already gone through them previously, but I would build upon that intuition that you have developed in the last section and then I will move slowly into probability distributions and then slowly into different types of probability distributions, getting into the details of continuous and discrete probability distributions, when do we use them, what are the real life experiences or examples of them and so on, right.

So let us begin with the section. Now here if you see, I have already told you about random event, random events are events wherein the outcomes are all random, right and all of those random events, we can attach a numerical value to it, which basically are called as random variables and the process itself is the randomization, right. So think of flipping a coin twice and you are looking for the number of heads, when you flip a coin twice, you can get only 4 outcomes, head, head, head, tail, tail, head, tail, tail, right.

So if you say the random variable is the number of heads that you get, so in that case you can get either 2 heads or 0 heads or 1 heads, that is the output, so random variable in this case can only take 3 different values, 0, 1 or 2 because you are only looking into the number of heads that you had. Similarly, think of if I say what is the likelihood that a random variable takes a value between a height of 5 to 6, there are various different scenarios, right, random variable in that case would be continuous, here in this case random variable is discrete because it is only taking specific discrete values. So these are the concepts that we will be developing upon but at least from the random variable point of view, you should be clear with it, random variable is a variable which takes any value randomly or any numerical value randomly.

So you do not know well in advance about which particular value it is going to take, the only thing that you will know is or basically the probability that it is going to assume, you only know is what are the outcomes that you have, so when you toss a coin 7 number of times, you only know the outcomes could be head or tail but you do not know how many heads would come, how many tails would come, what is the combination of heads and tails that you will be getting, right, so all of these things for these uncertainties or quantifying all of these uncertainties, we have scenarios like probability and we will be discussing about all of those things in details. Now if you look into, I got introduced you to the concept of discrete and continuous, discrete is when you have specific values, let us say when I say Britannia produces a number of biscuits everyday, they could produce 50 biscuits, they could produce 100 biscuits but they will never be producing 105 biscuits or let us say 0.5 biscuits, right, because those are not specific things, they are not whole numbers, when I say whole numbers, they are not exactly equals to 50 or 60 or 70 or 80 or 1000 and so on, they are in decimals, that is what is discrete, discrete is something which is specific, okay, similarly discrete can also be the number of heads that you have when you toss a coin 100 times and the number of balls that you have in the entire room full of balls or let us say the number of people that you have in the country, it cannot be a decimal number, right, so you cannot have the number of people living in a country which is up point or in decimals, right, similarly you also have on the other hand continuous random variables, continuous random variables are those which are uncountable, let us say height of people, now in between 5 to 6, there could be infinite number of people, height of 5 feet and height of 6 feet, so you have a range within which you can have infinite possibilities, when I say a range, a range could be as small as 5.1 to 5.2 and the range could be as big as 2 to 6, right, when we talk in terms of height, so continuous random variables can take any value within a continuous range and that is why they are called as something which can take infinite number of values, uncountable number of values, okay. Now, similarly when we talk about random variables, random variables since they can take many number of values, we have a number of values which can be thought of a distribution of values, right, so then comes how can we relate probabilities to each and every outcome that the random variable can take, so for example when you toss a coin 2 times, you have only 2 outcomes, let us say head and tail but you can have combination of head and tails, you can have head and head in one, head and tail in one, tail and tail in one and tail and head in one, right, so in that particular case if you think of the outcomes whether probability of getting 2 heads or probability of getting 1 head, you can attach probabilities to each and every outcome that you have, right, so if you toss a coin twice, you have at max 2 heads that you can get, so you can can attach probability of getting 0 head, probability of getting 1 head, probability of getting 2 heads and then you can sketch out the entire distribution.

So that's the concept behind distribution. Similarly when I say a random variable which is about weights of people living in India then you can sketch the entire probability distribution of what is the probability that any particular individual chosen randomly has a weight of let's say greater than 7 or lesser than greater than 70 or lesser than 20 or any specific value for that matter. So the idea is for the entire range of values that the random variable can assume if we can attach probabilities to it.

So probability distribution is nothing but a process of attaching probabilities to entire range of values that random variable can attain. So in that case we have different types of probability distribution because we have two different types of random variables which are discrete and continuous. Hence the number of probability distribution or type of probability distributions are also two types which is one continuous probability distribution the other one is discrete probability distribution.

That's what we are going to learn but before that if you think of the outcomes of any random experiment it has a specific number of outcomes. You cannot have multiple infinite outcomes. So when you toss a coin or when you roll a dice you have a specific number of outcomes.

Now in that particular case the probabilities that you have let's say when you roll a dice multiple number of times the numbers that you get on different different faces on the top faces of the die you could get 2 and 3 you could get 4 and 5 and 2 dice rolled and so on. But if you are trying to get to the probabilities that the die can or the two trials of the die can assume you would always see that the total probability of all the events or all the outcomes of the random event would always be equals to 1. So that's what is one of the golden rules that we sketch out. So in the entire distribution of values which random variable can assume the probabilities attached to each and every value would first sum up to always 1 second the probabilities can never be negative.

The probabilities either would be greater than 0 and less than 1 but they would never be negative. You cannot say that the probability of getting a head and a coin toss is negative 0.05. So that's one of the things. Now coming back to different types of discrete distributions.

So we talked about two different types of distributions. One was discrete and the other one was continuous. In discrete distributions we are going to study about two specific type of discrete distributions.

One is binomial the other one is Poisson. So let's get started with binomial. Binomial is a special distribution wherein you are counting successes in fixed number of trials.

When I say fixed number of trials what I mean is think of when you toss a coin the outcomes of tossing a coin is only two head or tail. So you have tossed a coin one number of time and the outcome that you have got is getting head or tail. Similarly you could toss a coin multiple number of times.

But the outcome would still be the same either getting head or tail. Now when you toss a coin the very first time versus when you toss the coin the second time both of these are independent events. So second time when you're tossing a coin the probability of getting head doesn't get influenced by what you have got the very last time.

So all of these things when we talk about in terms of binomial distribution the trials that we are doing they always have to be independent the trials also have to be fixed. Let's say if I want to know out of 100 flips of coins that I do what is the probability of getting 70 heads in that case the number of trials total number of trials is 100 right and we already know the probability of success which is probability of getting a head it's 0.5 and 0.5 because in one trial you can only get one head and one tail out of which both of both of them has equal probabilities of 0.5 right and if you think of they are independent trials no trial is dependent on each other. So binomial distribution is a specific or special type of distribution wherein all of these properties should hold true so that we can qualify something as a binomial distribution.

So what are these properties these are four conditions these four conditions are called as bins the very first thing is binary the outcome has to be only two either success or failures or no clicked or not clicked the trials has to be independent let's say for example the outcome of one trial doesn't affect the next trial a fixed number of trials has to be there I told you that we are tossing a coin only 100 times 500 times 700 times but you cannot do that trial infinite number of times right actually you can do but then it results in normal distribution which is what we learn later in the section. The success the probability of success is already known okay when you have all of these things defined we have a nice looking formula which is called as probability mass function or a function which attaches the output probabilities to number of trials that you are doing so let's say for example if you choose here and choose k p to the power k 1 minus p whole to the power n minus k what does it mean it means that how many ways you can get to a success so let's say I define success as getting heads or the example that I took right I can toss coin 100 times and while tossing a coin the success for me is to getting head so let's say what is the probability of getting 70 heads out of 100 trials now probability of success in this case of probability of getting head is 0.5 that would be 0.5 in any trial that you will be doing the second thing is if you think of how many ways we can choose 70 heads out of 100 trials it could be 100 choose 70 and that's what is the first part of the formula second part of the formula which is p to the power k is nothing but the probability of 70 successes because in this particular example which I took k was 70 right similarly probability of failure is n minus k p 1 minus p whole to the power n minus k which is basically about the probability of getting tail because 1 minus p if we think of p as the probability of getting head 1 minus p would be the probability of getting tail right so p is a probability of success 1 minus p is a probability of failure and this is the entire nice looking formula using which you can sketch out the total probability that you would get or let's say how what is the probability of getting 7 heads out of 10 trials or 100 heads out of 500 trials and so on you can also think of various different things let's say the what is the probability of getting 10 defective bulbs when you have 100 sample of bulbs right if you already know the probability of detecting defective bulb and you have total number of sample which is the number of trials as 100 and you already know the probability of success in this case which is defective you can easily sketch out what would be the probability of getting 10 defective bulbs or 50 defective bulbs and so on similarly you can get to probability of getting 100 good biscuits let's say you are producing a lot of biscuits now the outcome could be either good biscuits or bad biscuits right the outcomes are only two so in this particular case in binomial distribution by means two and we are only discussing about two outcomes right so binomial distribution in this particular case when we discuss about the number of biscuits being produced and we would want to know how many good biscuits would be produced and we already know what is the probability of getting or producing good biscuits then we can sketch out the probability of producing 100 good biscuits versus 200 or 1000 and so on all of these things can be easily found out using the simple looking formula which is nothing but the probability of selection of certain number of things in a fixed trial then probability of success and probability of failure so that's what it is now on in the bottom what I also have here is you can look into number of trials let's say total number of trials are 10 okay that's what you have it has been sketched out to and you see that what is up a probability of success up to one trial two trials three four five and so on you see the probabilities now if you will add the probabilities all of them up till 10 it would add up to 1 you see the probability mass function on the left hand side which is the y value number of successes on the x axis now once I change the number of trials look how does the entire distribution changes right and when I change the probability of success as well how does that formula comes into effect and the entire distribution of probabilities changes now in this particular case the number of successes is still limited what we are changing right now is the the probability of success that's all so if you think this is what what happens when you play around with the formula and change either the value of n or the value of p okay so that's that's what I have visualized or I have tried visualizing so when you increase more and more number of trials you will see that your output would actually assume a normal looking curve which is a bell-shaped curve I will tell you about it in details in later section but at least you got to know about what exactly it is right now we also have a specific or special type of distribution which is also discrete which is called as poison now a very good example that I can tell you is say that the total number of messages on your phone on any given day are hundred now if you would want to find out how many messages would come tomorrow in that case you can use poison distribution to find that probability what if I say that the total number of calls that I get on a particular day is 15 and I would want to estimate the total number of calls which would come tomorrow right so poison distributions are actually special types of distribution wherein you're counting events within an interval the interval in this particular case which the example that I gave you about was the number of phone calls in a day 24 hours interval similarly let's say within 15 minutes of window the probability or not let's not get into probability first but let's think about it so let's say I say there are four buses which comes at a bus stop in every 15 minutes now you know that the average number of buses in every 15 minutes is 4 okay so you have two informations one average and then the interval if you have these two informations you can sketch about probabilities of how many number of bus could come in in any interval okay so that is the importance of poison so let's say you run a bakery shop and you know that the total number of customers that would come on your shop on any given day is on an average 100 now if you would want to know because let's say there is a festival tomorrow you would want to know that what is the possibility that 200 people would come okay that's a rare event but still you can approximate the probability that 200 people would a probability of 200 people coming to your shop because you already know about two specific things one the interval in a day 200 people or 100 people are coming on an average second you have the defined interval so this portion distribution works only out of these two contexts okay now if you see here I have a probability mass function which is nothing but a formulation which attaches probability to each and every output so if you see here the is very simple e to the power minus lambda lambda to power K by K factorial let me help you understand or sketch this on a board so what what we are saying we are saying that let's say let's assume this example number of calls calls per day is equals to 15 okay you have been given the idea about average now you would want to know what is the probability of getting 20 calls okay so the formula for that is e to the power minus lambda lambda to the power X by X factorial or the probability mass function when I say formula what you have to assume is I am referring referring to probability mass function so in this particular case lambda is already given as 20 sorry lambda is already given as 15 that's what your average value is so we'll just feed it into the formula which is e to the power minus 20 sorry e to the power minus 15 and 15 to the power 20 divided by 20 factorial so you have everything required to calculate the output you have everything here I have slightly make it look not so good so let me rewrite it e to the power minus 15 15 to the power 20 by 20 factorial so this is what it is so you can easily calculate the output and get to know about what is the probability of getting 20 calls and why it is important think of why this particular distribution is very important so let's say you are running a call center and the average number of calls any call center employee in your company receives is let's say 80 okay and you would want to estimate total number of calls which would be which is received by any of those call center employees in the next two hours of time on next three hours of time and all of these things can be easily estimated using Poisson distribution so it becomes really really important because you can plan you can scope out things you can discover different types of things which are about to happen which might happen and so on and this distribution is really really effective when we talk about rare events anyways okay so this is second type of discrete distribution that I would I wanted all of us to learn about now we will quickly move into one of the most important distributions which is called as normal distribution and this distribution is continuous distribution so let me go to the board and let me go here so let's say if I sketch out the heights of people living in India so let's say on my x-axis it's it is heights okay and heights could start from literally zero and it could may go up to 10 feet but if you would have to plan the total number of people with a specific height let's say in between 0 to 1 there might be very few and then in between 1 to 2 there might be a little more 2 to 3 there might be a little more than 3 to 4 a lot more right and then 4 to 5 and then so on so this is how it might look like it might look like this normal looking curve so basically what I'm trying to tell you is you might sketch out a distribution which exactly looks like this okay now you see that this is a continuous set of values so let's say if I say that what is the probability of on the y-axis I have the probabilities and on the x-axis I have different height values that any individual can attain so you can easily find out what is the probability of an individual having a height of or having a height in between 4 to 5 so let's say these are two different ranges 4 to 5 and in between this what is the probability that if we have to find out what is the probability that I randomly choose an individual has a height 4 greater than equals to 4 and less than equals to 5 all of these types of questions could be easily answered via normal distribution okay and any continuous distribution for that matter provided that the variable that we are talking about it follows that distribution so generally what happens is you would get to know about central limit theorem and there are also law of large number and different types of things which governs why do we use normal distribution every single time but think of any physical quantity height weight pressure or any any other thing right temperature you would get normal looking looking curve in all of these things okay normal distribution for that matter is a probability distribution wherein the values that the random variable can take is infinite when I say infinite if you think of and on my board when I had sketched out the probability of taking any value you can have two different intervals let's say 4 and 5 and in between that you can have infinite number of values similarly you could have a very small interval as well you can have let's say 4.1 and 4.5 in between that also you can have infinite number of values and that work that's what is continuous random variable now when we take probabilities of all the range of values that the entire random variable can take we are talking about probability density function of a continuous distribution so a function which takes all the amount of continuous probabilities or let's say all the probabilities that the continuous variable can take is continuous distribution and within which we are specifically discussing about normal distribution now normal distribution is this sort of distribution which looks a bell like a bell shaped curve and normal distribution is governed by two different parameters so in this particular case the parameters of normal distribution are two things one the mean of the distribution second the standard deviation okay now if you if you look into I can easily sketch out the mean could be somewhere around here the middle line that I am talking about right this could be very well be the mean so mu in this case I am writing that as mean and standard deviation is the stretch from that mean so let's say one standard deviation away how what how many values are there so the shaded region is that one standard deviation away in general up to one standard deviation away there are 66 or 67 percent of the value that lies in a normal distribution up to one standard deviation away similarly up to two standard deviation away we have 95 percent of the total values that lie so if you think of it would seem like this or let me color it in a different color so that you can understand so this is what what I am talking about okay now similarly threes up to three standard deviation away we have ninety nine point seven percent of the total values that can be under this umbrella of normal distribution okay so this is these are very specific properties of normal distribution but if you if you think of the formula for normal distribution or the formula which I say as probability density function now firstly let's look into all of these probability functions right so there are two different types of distribution that we know so probability functions can be of two types for two the two types of distribution that we know one is called as probability mass function the other one is called as probability distribution function this is mass function because it attaches mass or weights like thing between the actual or exactly and those observations so what I am talking about is that say you toss a coin when you toss a coin there are only two outcomes head and tail now if I sketch out the probability mass function for a tossing a coin you would have only two outcomes one head the other one is tail or let's say head is zero and tail is one so in that case I have the probabilities are exactly 0.5 for both of them right so I can draw a bar which are at 0.5 okay so this is this is exactly the probability mass function of of the the trial that we did similarly if I am thinking about drawing what is the height of people living in India in that case my entire distribution might look like this now why is this continuous and in the case of discrete distribution you saw that in between 0 and 1 there was nothing in between head and tail there was no say nothing because a coin toss can never result in an intermediate of result like you cannot have anything in between 0 and and 1 you cannot have anything between head and tails.

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